ECO101: Introduction to Microeconomics
Assignment 01| Spring 2025
Answer All Questions | Total Marks – 35 | Deadline - March 09, 2025, by 11 am sharp

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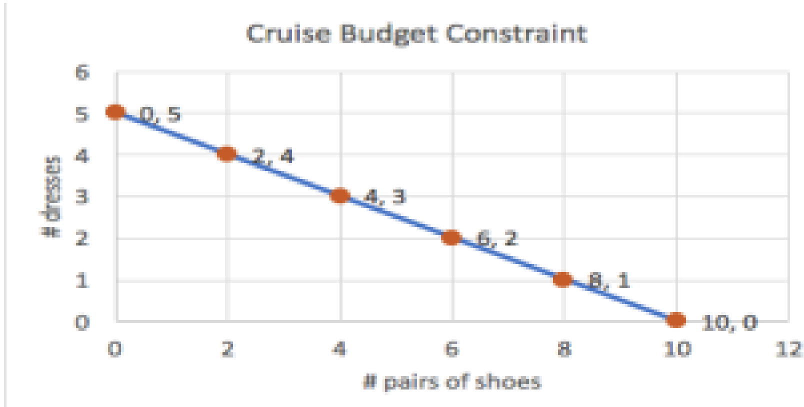
Instructions:

- *You must submit this assignment as a group. Each group member is expected to contribute to the answers. No individual assignments will be accepted.*
- *Each group member will receive the same score, which is the total score obtained in the overall assignment.*
- *Submit your answers to the assignment in clear A4 size papers, which should be stapled properly. Loose papers will not be accepted.*
- *Answer the following questions in the given order only. Make sure to number your answers clearly.*
- *Do not rewrite, copy, or attach the question paper with your answers.*
- *Both handwritten and printed assignments will be accepted. Graphs must be drawn by hand.*
- *The assignment must have each member's full name, ID, and section number on the first page.*
- *Any assignment submitted after the deadline will be considered late.*
- *Late submissions will only be accepted on the immediate next class after the deadline, and there will be a deduction to the score. No reminders will be given for deadlines and late submissions.*
- *Late Assignment Deduction: You will only receive 20% of the obtained score.*

Question 01

	Cars	Bikes
Country A	10	5
Country B	8	2
Country C	12	4

(a) Which countries have the absolute advantage in producing cars and bikes? **(1 mark)**
(b) What is Country B's opportunity cost of producing 10 cars? **(1 mark)**
(c) Showing the calculation of the opportunity cost for producing both the goods for each country, explain which country has a comparative advantage in producing cars and which country in producing bikes? **(4 marks)**
(d) The graph below represents Alice's budget constraint:



Alice is planning a cruise to the Caribbeans and has a budget of $500. The price for a pair of shoes is $50, while the price for a dress is $100. She found 6 pairs of shoes and 2 dresses she likes. She decides that she does need one more dress. What is the opportunity cost of buying a third dress for Alice? **(2 marks)**

Question 02

For each of the following scenarios, use a supply and demand diagram to illustrate the effect of the given scenario on the equilibrium price and quantity in the specified competitive market. Explain whether there is a shift in the demand curve, the supply curve, both, or neither.

(a) China and Mexico are major producers of textiles. Workers in Mexico decide to go on strike. Show the effect on the market for Mexican textiles. **(2 marks)**

(b) Show the effect of the situation described in (a) on the market for Chinese textiles. **(2 marks)**

(c) KFC and BFC both sell crispy fried chicken, each using their own secret recipe. Both sell their fried chicken at Tk 210. Due to the recent drop in sales, KFC has decided to lower their price to Tk 195. Explain and illustrate the effects on the equilibrium price and quantity of BFC's fried chicken. **(2 marks)**

(d) An unexpected temporary heat wave hits Bangladesh. At the same time, the government introduces a tax on ice cream, which is paid by producers. It is stated that the effect of the change in the market for ice cream due to the heat wave is greater than the effect of the change due to the tax imposed on the ice cream producers. Explain and illustrate the effects on the equilibrium price and quantity of ice cream. **(3 marks)**

(e) The local production of mangoes has increased significantly. This has coincided with a decrease in consumers' income. Assume that mangoes are an inferior good. Given that the effect of the decrease in consumers' income is lower than the effect of the increase in the production of mangoes. Explain and illustrate the effects on the equilibrium price and quantity of mangoes. **(3 marks)**

(f) Imagine that the demand for Bangladesh's national flag has increased due to an upcoming Cricket match. At the same time, there is a dock workers' strike going on at the Chittagong seaport and flag manufacturers have very little supply of the raw material needed to make these flags. Draw a demand-supply diagram to show what happens in the market for Bangladeshi flags, given that the demand effect is more significant. **(3 marks)**

Question 03

Given, the following demand, $Q_D = 40 – 2P$ and supply, $Q_S = 2 + P$ equations:

(a) Calculate the equilibrium price and quantity. **(1 mark).**

(b) If the government imposes and fixes the price at BDT 12, will there be a shortage or a surplus? Calculate the amount of shortage or surplus. **(3 marks)**

(c) Provide a graphical representation of your answers from (a) and (b). **(2 marks)**

(d) Consider the market for butter for parts (i) and (ii) below.
The demand curve is given by $Q_D = 300 - 2P + 4I$, where I is the average income. The supply curve is $Qs = 3P - 25$�� $- 25$, where is the price of milk. ������
(i) If the average income in Cambridge is $I = 25$ and the price of milk is �� = 1, what is �� the market clearing price and quantity in Cambridge? **(2 marks)**
(ii) Suppose that bad weather conditions raise the price of milk to �� = 2. Find the new �� equilibrium price and quantity of butter in Cambridge. **(2 marks)**
(iii) Draw a graph of the market for butter to illustrate your answers in parts (i) and (ii). **(2 marks)**